

RSU-101 Hardware Guide

HW Version: 0B

V1.1

Table of Contents

1. RSU at a Glance 1

2. Accessories List 3

3. Mechanical Drawings 7

4. Pin Definition 8

5. Cable Gland 11

6. Surge Protector (Optional) 13

7. Mounting Bracket 17

8. Installation Sequence 22

1. RSU at a Glance

Front View:

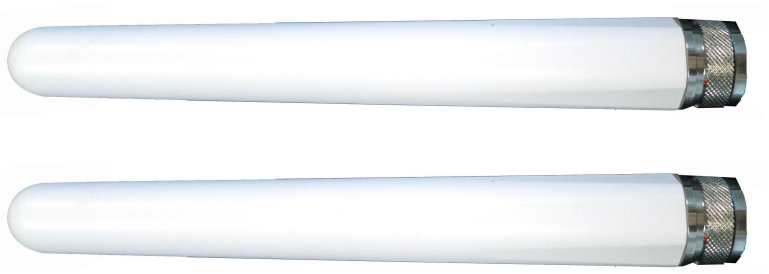


Rear View:



2. Accessories List

1. DSRC antenna x2



2. GNSS antenna x1



3. Cable gland x2



4. Power cable gland x1



5. Mounting bracket x1



6. Clamp x2



7. M6 screw x4



8. Spring washer x4



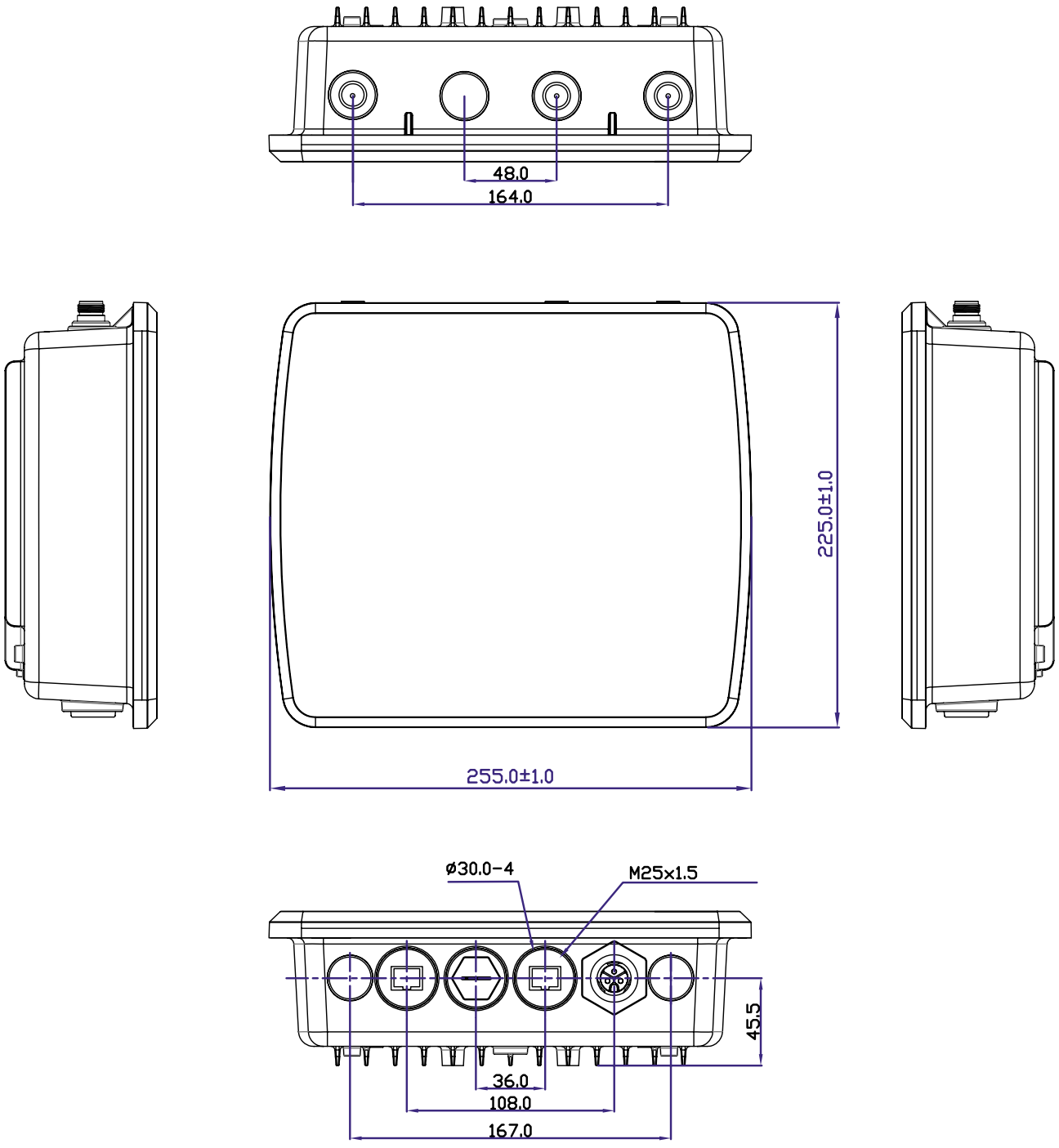
9. Flat washer x4



10. Plastic anchor x4 , Pointed screw x4

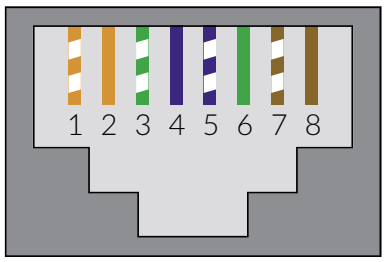


3. Mechanical Drawings



4. Pin Definition

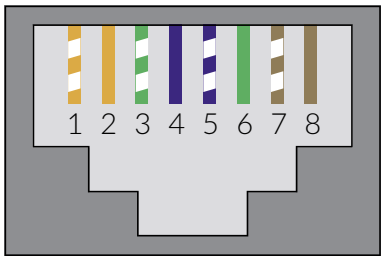
RJ45 Port1 (RS-232 Console/Ext GNSS, CAN, GPIO)



T568B

Port	Pin No.	Color	Pin Name	Type	Description
Console / Ext GNSS (EIA/TIA-561)	2		1PPS	I	GPS Time Pulse input (3.3V)
	5		RXD	O	Serial data to DTE
	6		TXD	I	Serial data from DTE
	4		GND	GND	Ground
GPIO	1		GPIO7	I/O	GPIO (3.3V) High-level output current: -10mA Low-level output current: 25mA
	3		GND	GND	Ground
CAN2	7		CAN_L	I/O	CAN Low (±40V)
	8		CAN_H	I/O	CAN High (±40V)

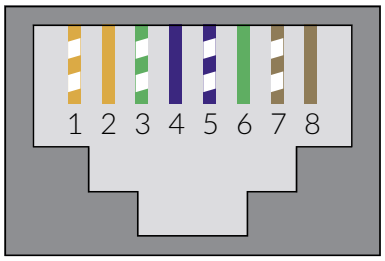
RJ45 Port2 (CAN, GPIO)



T568B

Port	Pin No.	Color	Pin Name	Type	Description
GPIO	1		GPIO2	I/O	GPIO (3.3V) High-level output current: -10mA Low-level output current: 25mA
	2		GPIO3	I/O	
	3		GPIO4	I/O	
	4		GPIO5	I/O	
	5		GPIO6	I/O	
	6		GND	GND	Ground
CAN1	7		CAN_L	I/O	CAN Low (±40V)
	8		CAN_H	I/O	CAN High (±40V)

RJ45 Port3 (Power over Ethernet, PoE)



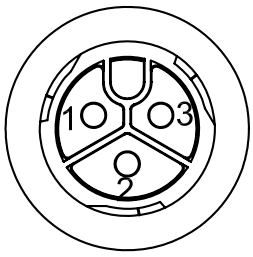
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PoE+ 802.3at (25.5W)

NOTE: For safety concern, please do not connect PoE injector and Power Input at the same time. Always connect ONLY ONE power source.

Pin No.	Pin Name	Type	Description
1	TX+	I/O	Transmit +
2	TX-	I/O	Transmit -
3	RX+	I/O	Receive +
4	VDC+	PWR	PoE Power
5	VDC+	PWR	PoE Power
6	RX-	I/O	Receive -
7	VDC-	PWR	PoE Power
8	VDC-	PWR	PoE Power

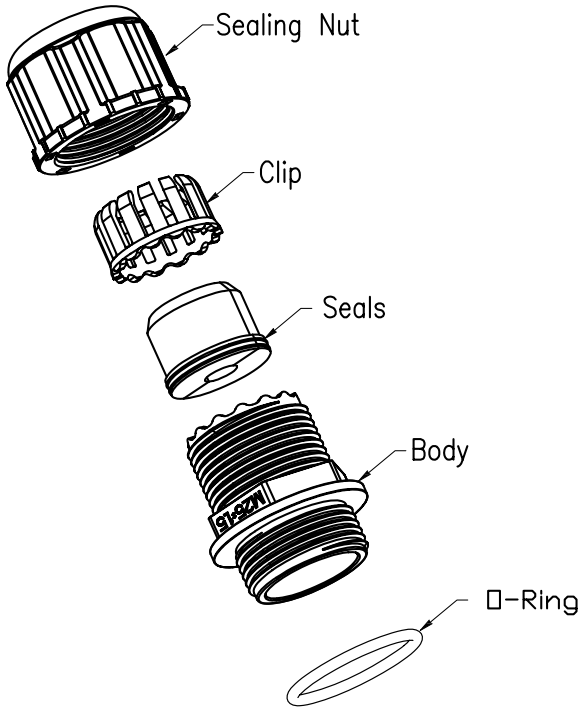
Power Input



Pin No.	Pin Name	Level	Description
1	PWR	6-32V	System main power
2	GND	-	Ground
3	Vbatt	6-32V	GNSS RTC power

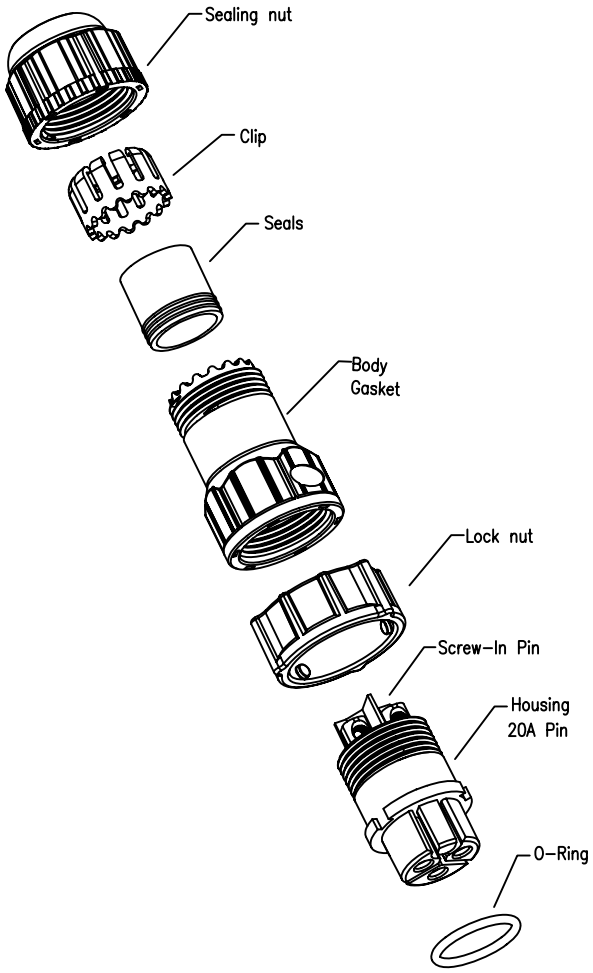
5. Cable Gland

Cable Gland for RJ45 Cable:



1. Cable diameter: 4.5-6.5mm.
2. Insert RJ45 cable through sealing nut, clip, body and O-ring in a row. (see above drawing for parts direction)
3. Securely connect the RJ45 connector with the RJ45 port.
4. Fasten the body to the RJ45 port, with O-ring side facing the metal housing.
5. Sit the seals firmly into the body.
6. Push the clip and seals all the way down to the body.
7. Fasten sealing nut with the body.

Cable Gland for Power Cable:



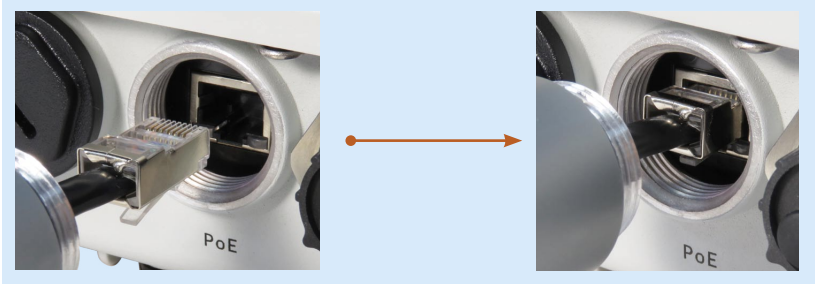
1. Cable diameter: 5.5-8.0mm.
2. Conductor: 14-18AWG.
3. Insert power cable through sealing nut, clip, seal, body gasket, and lock nut in a row. (see above drawing for parts direction)
4. Connect the conductors and screw-in pins. Fasten screws to ensure strong connection
5. Fasten body gasket with housing
6. Sit the seal firmly inside the body gasket.
7. Push the clip and seals all the way down to the body.
8. Fasten sealing nut against the body.

Note: the sealing nut can only be fastened ONCE. A sprocket mechanism inside the sealing nut will keep it from loosening once fastened.

6. Surge Protector (Optional)

PoE Surge Protector:



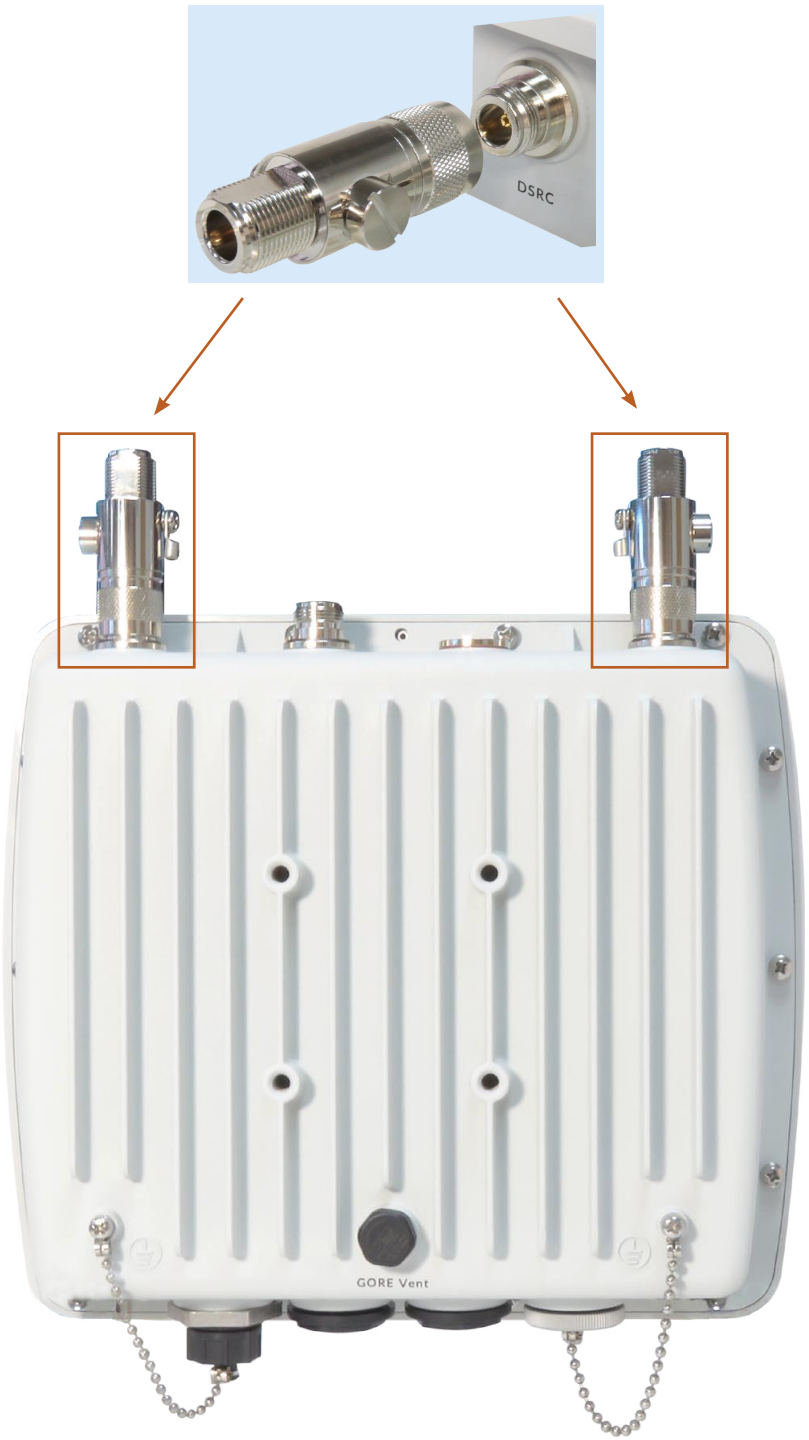


Breakdown Voltage	57V
Max. Surge protection	±6KV (1.2/50uS), 3KA (8/20uS)
Max. ESD protection	±15KV (air), ±8KV (contact)
Pins Protected	Data/PoE : 1/2, 3/6, 4/5, 7/8 all pins
Data Rate	10/100/1000 Mbps
Response Time	< 1ns
Operating Temperature	-20 ~ +75°C
Storage Temperature	-40 ~ +85°C
Operating Humidity	0% ~ 95% non condensing

1. Connect RJ45 cable inside the surge protector to PoE Port
2. Fasten the surge protector to the housing.
3. Connect the RJ45 cable to the surge protector as described in chapter 5: Cable Gland for RJ45 Cable.

Note: surge protector is not included in standard package.

Antenna Surge Arrester:

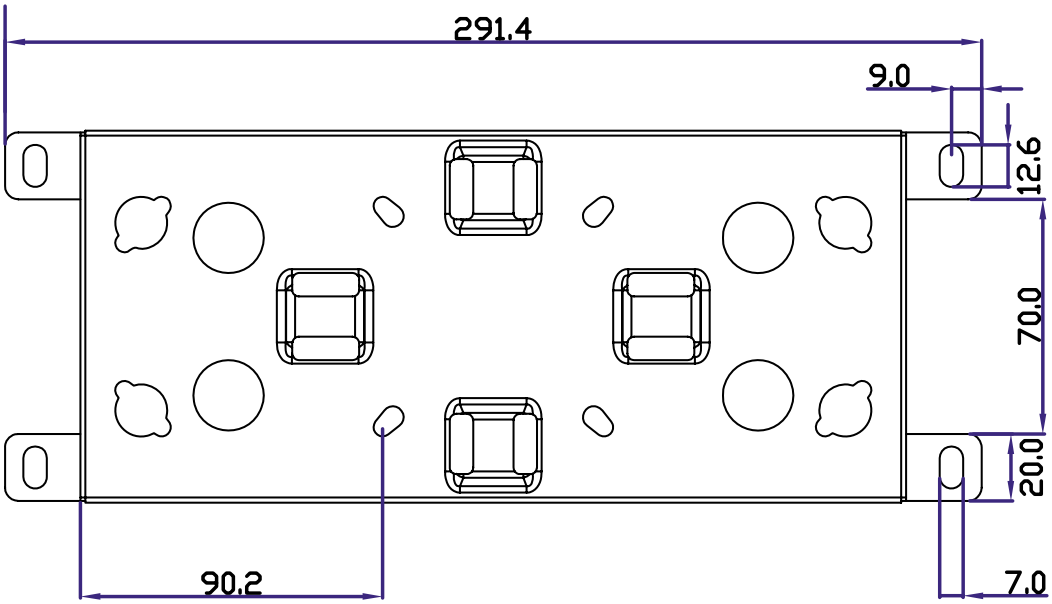


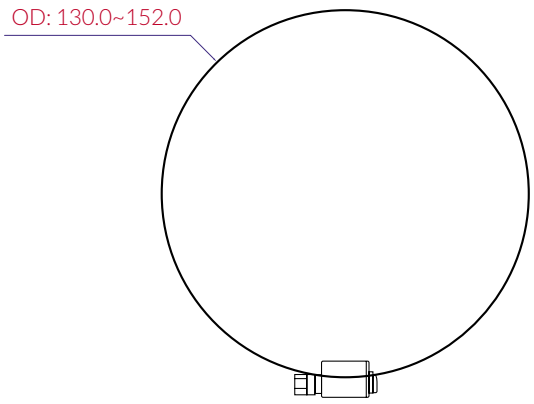
VSWR	3~6GHz 1.5(max):1
Insertion Loss	1.2dB(max)@ 6GHz
Gas Tube DC Breakdown Voltage	90V ± 20%
Impulse Discharge Voltage	5KA min (wave 8/20 μs), 10KV min (wave 1.2/50 μs)
Insulation Resistance	DC 50V > 10,000MΩ

1. Connect surge arresters to the DSRC antenna ports.
2. Connect DSRC antenna to the surge arresters.

Note: surge arresters are not included in standard package.

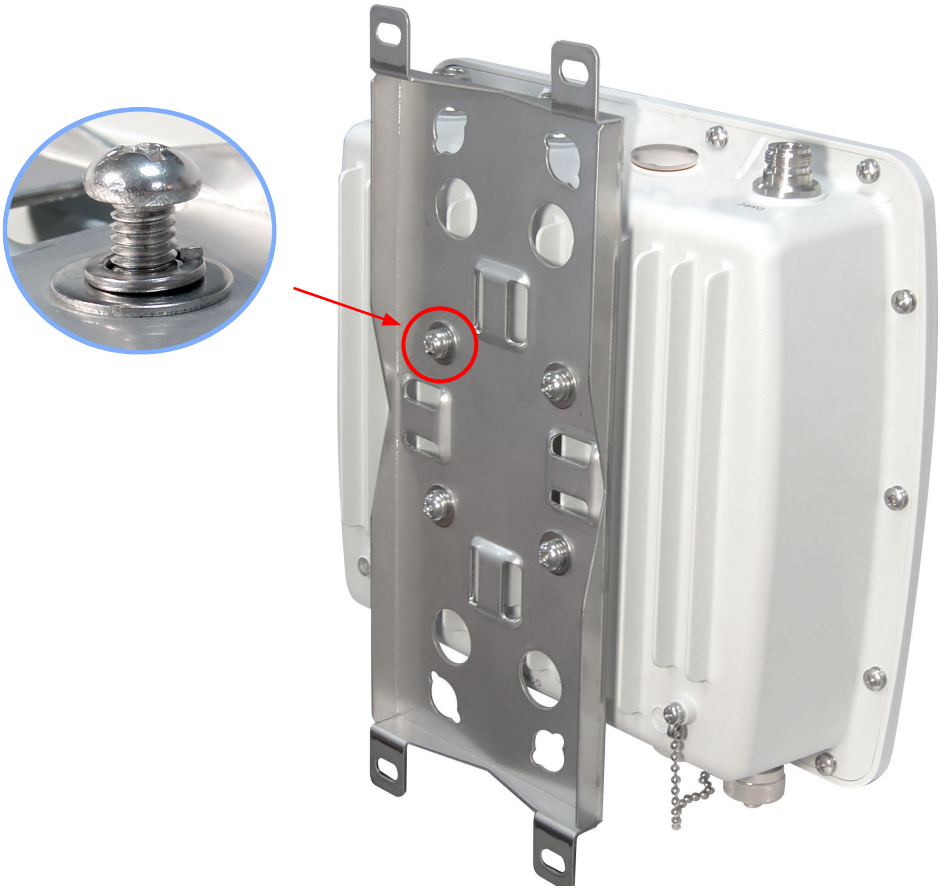
7. Mounting Bracket



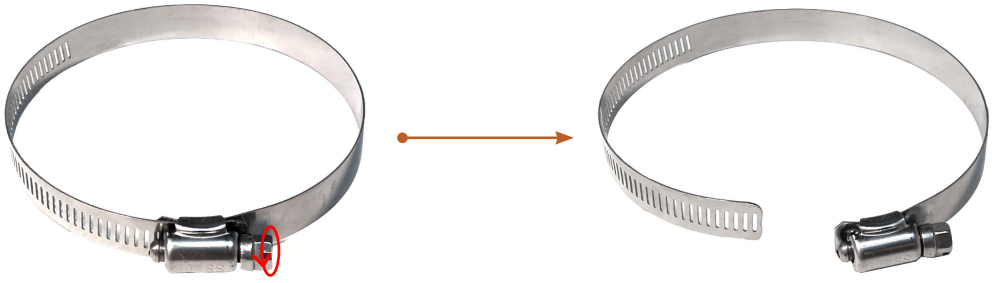


Mounting on a Poll:

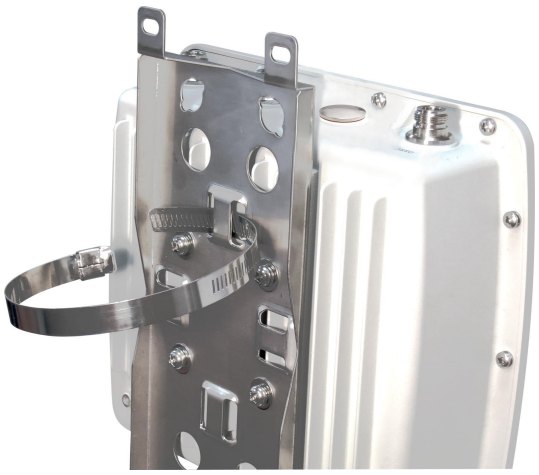
1. Assemble the bracket on the housing. (M6 Screws x4, be sure to use a spring washer and a flat washer for each screw)



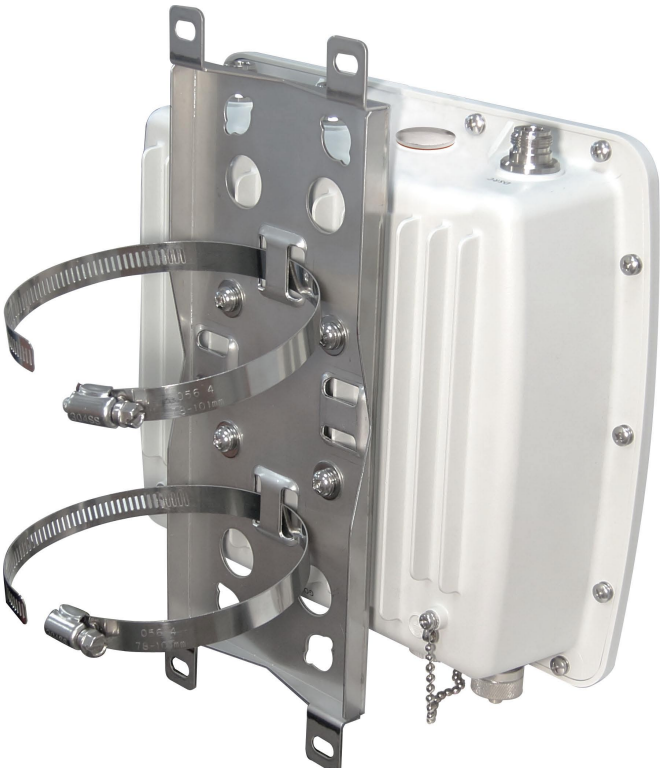
2. Loosen the screw on the clamp to open it



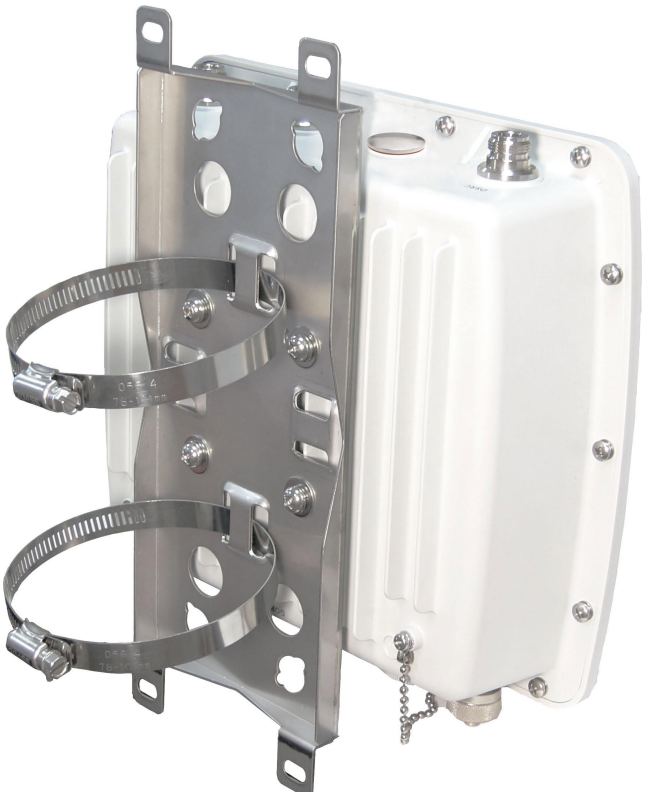
3. Insert the clamp into the slits on the bracket



4. Wrap the clamp around a poll, and fasten the screw to close the clamp (Poll Diameter: 130~152mm)



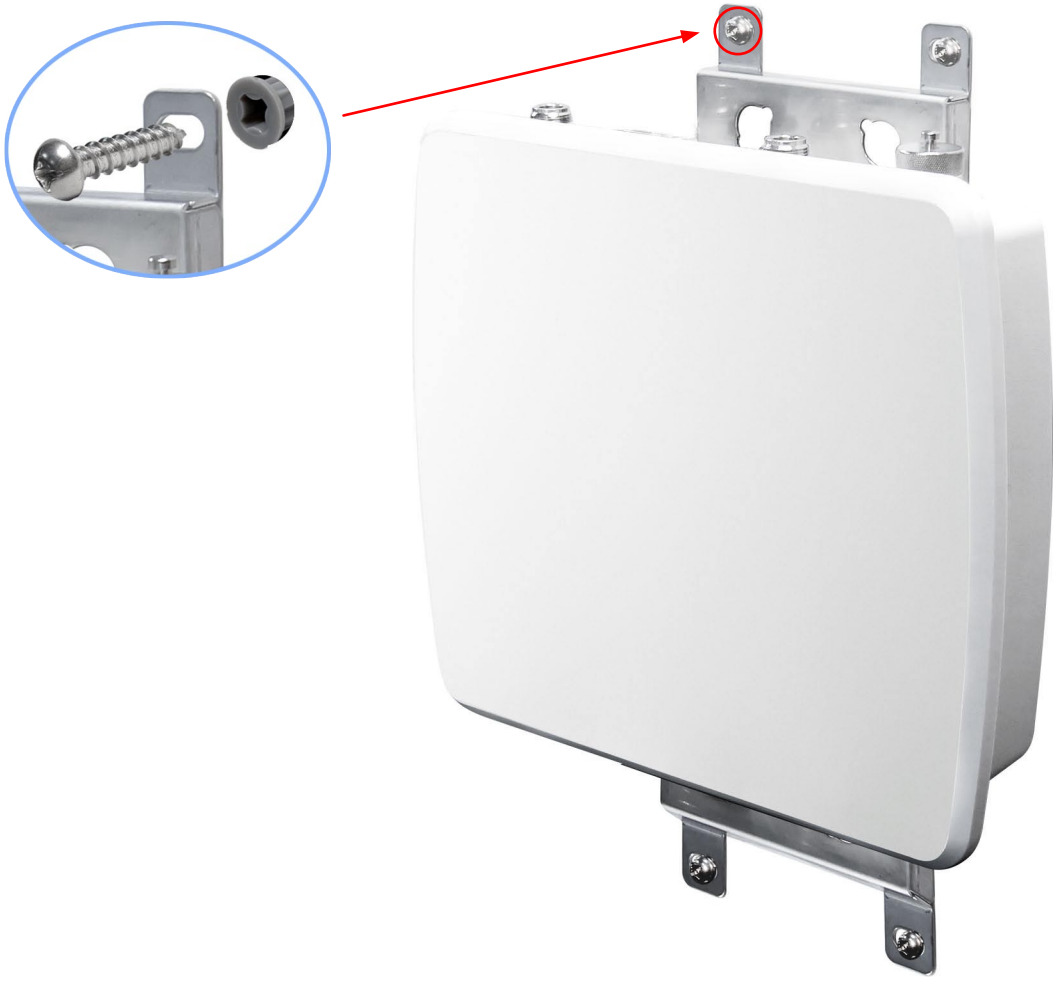
5. Fasten the screw on the clamp until RSU is mounted securely.



Mounting on a Wall:

1. Assemble the bracket on the housing. (M6 screws x4, be sure to use a spring washer and a flat washer for each screw)

2. Use anchors to mount the RSU on a wall. (plastic anchor x4, pointed screw x4)



8. Installation Sequence

To avoid interference, it is recommended to follow the installation sequence:

1. Mounting bracket
2. Cable glands
3. Antennas

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